Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	High-Performance Soft Magnetic Cores via Screen Printing			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery (), Advanced Mobility (), Next-gen Nuclear Power ( ), Advanced Bio (), Aerospace/Marine (), Hydrogen ( ), Cybersecurity ( ), AI (), Next-gen Communication (), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical - Materials		Electrical – Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Metal Materials	Functional Materials	Electrical and Electronic Components	Magnetic Material Components
Concept of Proposed Technology	The proposed technology enables additive manufacturing of high-performance soft magnetic motor components by screen printing and stacking thin magnetic sheets into near-net-shape substacks. In contrast to conventional punching and stacking of rolled electrical steel, the sheet thickness is defined by the printing process, which reduces material waste and avoids edge-related magnetic degradation caused by mechanical cutting. Electrical insulation between sheets is realized by printing a thin ceramic insulation layer, resulting in an automated manufacturing route for metal–ceramic layered composite components. This approach is particularly attractive for compact or small motor concepts, where conventional mechanical processing steps are cost-intensive and can deteriorate magnetic performance.			
Rationale for Joint R&D Project	To unlock the full potential of this manufacturing approach, joint R&D is required to (i) ensure uniformity and durability of printed substacks and insulation layers, (ii) validate electromagnetic performance through systematic testing, and (iii) transfer the material and component data into application-oriented design tools. The magnetic properties obtained from printed sheets and substacks can be used as input for motor simulation. Key manufacturing-related parameters such as the achievable stacking factor and the performance of full substacks in a motor-relevant environment			

	are not yet sufficiently established. Therefore, prototype components and a demonstrator motor should be built to evaluate real performance and enable a fair comparison with state-of-the-art laminated cores. Further advancements may include the integration of cooling channels, topology-optimized geometries, and locally adapted compositions through multi-material printing. Combining the expertise of a motor designer and materials/process developers will enable the validation of innovative soft magnetic components in relevant applications and provide a strong basis for long-term international cooperation with Korean partners.
Key Words	1. Soft magnetic components 2. High frequency 3. Additive manufacturing (additive screen printing) 4. Metal–ceramic layered composites 5. Powder metallurgy

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	The objective of this joint R&D project is to further develop the existing screen-printing approach for laminated soft magnetic cores based on conventional Fe–Si electrical steels and extend it to (i) high magnetic induction materials and (ii) high-frequency materials. For each material system, the printed insulation layer must be tailored to maximize electrical insulation while maintaining mechanical integrity during debinding and sintering. The project will therefore identify suitable powders, adapt paste rheology and printing parameters, and develop robust debinding and sintering schedules. Densification-induced shrinkage will be quantified and incorporated into component design (e.g., screen pattern/layout compensation). As a result, customized soft magnetic cores for high-frequency applications will be manufactured and characterized with respect to microstructure, mechanical integrity, and magnetic performance.			
R&D Target Specifications	Target specifications will be defined jointly with the industrial partner based on the intended application and operating conditions. Based on printed sheet trials achieving **≥ 99% density**, a target sheet thickness in the range of **100–200 µm** is envisaged for high-frequency cores. A **stacking factor ≥ 85%** should be achieved for multi-sheet substacks. Shrinkage during heat treatment shall be homogeneous, and the insulation layer shall remain intact without delamination or cracking.			
R&D Plan	The following project plan is proposed for a duration of 36 months and follows a structured approach covering component definition, process development, component manufacturing, and validation under application-relevant conditions. Phase 1: Definition of requirements (Months 1–3) Together with the project partner, target applications, key performance indicators (KPIs), geometrical constraints, and operating conditions will be defined. Based on these inputs, suitable material systems will be selected. Phase 2: Screen printing and evaluation of test geometries (Months 4–15) Powders and paste formulations will be screened and printed into test geometries (e.g. toroids and single sheets). Printing parameters and paste rheology will be optimized. Samples will be characterized magnetically and mechanically, and microstructural analysis will be performed to correlate processing, structure, and properties. Phase 3: Soft magnetic component manufacturing (Months 16–30) Component manufacturing will be scaled to application-relevant geometries. Insulation layers will be adapted to the selected material system. Shrinkage behavior will be quantified and incorporated into screen and component design to achieve the required sheet thickness and stacking factor without delamination. Full components will be built by stacking substacks produced via additive screen printing. Phase 4: Validation under laboratory and application-relevant conditions (Months 30–36) Substacks and full components will be characterized under laboratory conditions and subsequently validated in a demonstrator setup. Magnetic characterization will support motor design and simulated versus measured performance will be evaluated under representative operating conditions. Key Milestones • M1 (Month 15): Materials and process parameters validated on single sheets and test geometries • M2 (Month 30): Prototype soft magnetic components available for validation in application-relevant conditions			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Fraunhofer IFAM has long-standing expertise in powder metallurgy and advanced coating technologies for functional materials. For more than two decades, the institute has developed and characterized soft magnetic materials and components, covering process development, materials design, and microstructural and functional characterization. Within the joint project, Fraunhofer IFAM will lead the development of screen-printing processes, paste and powder selection, debinding and sintering strategies, and the systematic characterization required to link processing, microstructure, and performance providing a strong foundation for industrial implementation together with the partner.
Expected R&R of Korean SME Partner	The Korean SME partner is expected to contribute application-driven requirements and define the relevant KPIs and boundary conditions for component design and validation. The partner may adapt or redesign the magnetic system to leverage the benefits of additive manufacturing (e.g., integration of cooling channels, geometry optimization, or multi-material concepts). The SME will support application-oriented testing and validation under relevant conditions, provide technical feedback for iterative optimization, and contribute resources such as materials, know-how, and personnel. Continuous collaboration and knowledge exchange will ensure alignment between component development and application needs.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The joint R&D project is expected to deliver additively manufactured soft magnetic cores with high performance and reduced losses, particularly for high-frequency operation. The project will establish a clear understanding of achievable magnetic performance and manufacturing parameters (e.g., stacking factor, insulation integrity, and robustness) in comparison with state-of-the-art laminated cores. Through systematic testing—ideally including validation under application-relevant conditions—the collaboration will enable design-relevant material data and performance limits to be considered early in motor development. In the longer term, sustainable manufacturing of advanced magnetic cores can contribute to more efficient electric energy conversion and support improved energy efficiency across electrified mobility and other power-electronics-related applications.

6. References

(1) Mix & Lindemann-Geipel et al. IEEE TTE (2023) 10.1109/TTE.2023.3298488

(2) Lindemann-Geipel et al. Powder Metallurgy (2024) DOI: 10.1177/00325899241247319

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization